

**LAPORAN PRAKTIKUM
APLIKASI BERBASIS PLATFORM**

EXERCISE JAVASCRIPT



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EXERCISE JS

Exercise

- Create a function that calculates only the odd values in a sequence of numbers, given the min and max sequence value as inputs. Example:
 - min = 1, max = 8, result = 16 (1+3+5+7)
 - min = 4, max = 15, result = 60 (5+7+9+11+13+15)
- Open your previous CSS exercise file, add two buttons that call a function when clicked:
 - First button: change the alignment of the text to center or right
 - Second button: change the background color of one of the box. When button is clicked again, the background color is back to original color



Code:

```
<!doctype html>

<html lang="id">

<head>

  <meta charset="utf-8" />

  <meta name="viewport" content="width=device-width,initial-
scale=1" />

  <title>Wireframe Exact Placement</title>

  <link
href="https://cdn.jsdelivrivr.net/npm/bootstrap@5.3.2/dist/css/bootstr
ap.min.css" rel="stylesheet">

  <style>

    .container { max-width: 820px; }

    .wf { border: 1px solid #111; box-sizing: border-box;
background:#fff; }
```

```
.logo {  
    aspect-ratio: 1/1;  
    min-height: 120px;  
    display: flex;  
    align-items: center;  
    justify-content: center;  
    font-weight: 700;  
}
```

```
.sidebar {  
    margin-top: 12px;  
    display: flex;  
    flex-direction: column;  
    gap: 8px;  
}
```

```
.sidebar .nav-item {  
    padding: 10px;  
    min-height: 44px;  
    display: flex;  
    align-items: center;  
    font-weight: 600;  
}
```

```
.banner {  
    min-height: 120px;  
    background: #e6e6e6;  
    display: flex;  
    align-items: center;  
    justify-content: center;  
    font-weight: 700;  
}
```

```
.sub-bars .bar {  
    height: 30px;
```

```
background: #f3f3f3;

border: 1px solid #111;

margin-bottom: 8px;

display: flex;

align-items: center;

padding-left: 12px;

font-weight: 600;

}


.content-top {

display: flex;

gap: 12px;

align-items: flex-start;

}

.text-block-top {

border: 1px solid #111;

padding: 10px;

min-height: 120px;

flex: 1;

text-align: left;

}

.image-box {

background: #ccc;

border: 1px solid #111;

width: 220px;

aspect-ratio: 1/1;

display: flex;

align-items: center;

justify-content: center;

font-weight: 700;

}


.image-box.alt-bg { background: #ffd54f; }


.content-below .line {
```

```

        height: 28px;

        border: 1px solid #111;

        background: #fff;

        margin-bottom: 8px;

        display: flex;

        align-items: center;

        padding-left: 10px;

        text-align: left;
    }

    .footer-box {
        border-top: 6px solid #111;

        padding: 10px;

        text-align: center;

        font-weight: 700;
    }

    .gap-vert { margin-bottom: 12px; }

    @media (max-width: 767.98px) {
        .logo { min-height: 90px; aspect-ratio: auto; }

        .banner { min-height: 90px; }

        .image-box { width: 160px; min-height: 160px; }
    }
</style>
</head>
<body>

    <div class="container mt-3 mb-4">

        <div class="row">

            <!-- KIRI: col-3 -->

            <div class="col-3">

                <!-- Logo -->

                <div class="wf logo">Logo</div>

                <!-- Sidebar vertikal tepat di bawah logo -->

```

```

<div class="sidebar mt-3">

    <div class="wf nav-item">Navigasi</div>

    <div class="wf nav-item">Link 2</div>

    <div class="wf nav-item">Link 3</div>

</div>
</div>

<!-- KANAN: col-9 -->
<div class="col-9">

    <!-- Banner/header (sejajar dengan logo) -->
    <div class="wf banner">Banner</div>

    <div class="sub-bars mt-3">

        <div class="bar">Sub Header Bar 1</div>

        <div class="bar">Sub Header Bar 2</div>

    </div>

    <div class="mt-3">

        <div class="content-top">

            <div id="textBlockTop" class="text-block-top">

                <div style="margin-bottom:8px;">Baris teks 1 (sejajar
kiri gambar)</div>

                <div style="margin-bottom:8px;">Baris teks 2 (sejajar
kiri gambar)</div>

                <div style="margin-bottom:8px;">Baris teks 3 (sejajar
kiri gambar)</div>

            </div>

            <div id="imageBox" class="image-box">Gambar</div>

        </div>

        <div id="contentBelow" class="content-below mt-3">

            <div class="line">Baris konten 4 (mengisi lebar penuh
col-9)</div>

            <div class="line">Baris konten 5</div>

            <div class="line">Baris konten 6</div>

            <div class="line">Baris konten 7</div>

```

```

</div>

<div class="mt-3 d-flex flex-column gap-2">
  <div class="d-flex gap-2 flex-wrap">
    <button id="btnToggleAlign" class="btn btn-sm btn-
primary" type="button">
      Toggle Align (Center / Right)
    </button>

    <button id="btnToggleImage" class="btn btn-sm btn-
secondary" type="button">
      Toggle Image Color
    </button>
  </div>

  <!-- fungsi sumOdd -->
  <div class="d-flex gap-2 align-items-center flex-wrap">
    <label class="mb-0">Min</label>

    <input id="inputMin" type="number" class="form-
control form-control-sm" style="width:100px" value="1">

    <label class="mb-0">Max</label>

    <input id="inputMax" type="number" class="form-
control form-control-sm" style="width:100px" value="8">

    <button id="btnCompute" class="btn btn-sm btn-
success" type="button">Hitung Jumlah Ganjil</button>
  </div>

  <!-- hasil -->
  <div id="resultArea" class="mt-2" style="min-
height:28px; font-weight:600;"></div>
</div>
</div>

<div class="mt-3">
  <div class="footer-box wf">Footer</div>
</div>
</div>

```

```

</div>

</div>

<script>

    /**
     * Menghitung jumlah bilangan ganjil dalam rentang [min, max]
     (inklusive).
     * Contoh: min=1, max=8 => 1+3+5+7 = 16
     */
    function sumOdd(min, max) {
        min = Math.trunc(Number(min));
        max = Math.trunc(Number(max));
        if (isNaN(min) || isNaN(max)) return 0;
        if (min > max) {
            var tmp = min; min = max; max = tmp;
        }
        var start = (min % 2 !== 0) ? min : min + 1;
        var sum = 0;
        for (var n = start; n <= max; n += 2) {
            sum += n;
        }
        return sum;
    }

    // --- Toggle alignment ---
    (function() {
        var alignState = 'center';
        var btnAlign = document.getElementById('btnToggleAlign');
        var textTop = document.getElementById('textBlockTop');
        var contentBelow = document.getElementById('contentBelow');

        btnAlign.addEventListener('click', function() {
            alignState = (alignState === 'center') ? 'right' :
'center';

            textTop.style.textAlign = alignState;
            var lines = contentBelow.querySelectorAll('.line');

```

```

        lines.forEach(function(el) {
            el.style.textAlign = alignState;
        });

        btnAlign.textContent = 'Toggle Align (Center / Right) -  
Sekarang: ' + alignState;
    });
})();

(function() {
    var btnImage = document.getElementById('btnToggleImage');
    var imageBox = document.getElementById('imageBox');

    btnImage.addEventListener('click', function() {
        imageBox.classList.toggle('alt-bg');
        var active = imageBox.classList.contains('alt-bg') ? 'ON' :  
'OFF';
        btnImage.textContent = 'Toggle Image Color - ' + active;
    });
})();

// --- Hook tombol hitung jumlah ganjil ---
(function() {
    var btnCompute = document.getElementById('btnCompute');
    var inputMin = document.getElementById('inputMin');
    var inputMax = document.getElementById('inputMax');
    var resultArea = document.getElementById('resultArea');

    btnCompute.addEventListener('click', function() {
        var min = parseInt(inputMin.value, 10);
        var max = parseInt(inputMax.value, 10);

        if (isNaN(min) || isNaN(max)) {
            resultArea.textContent = 'Masukkan nilai min dan max yang  
valid.';
            return;
        }
    });
})();

```

```
}

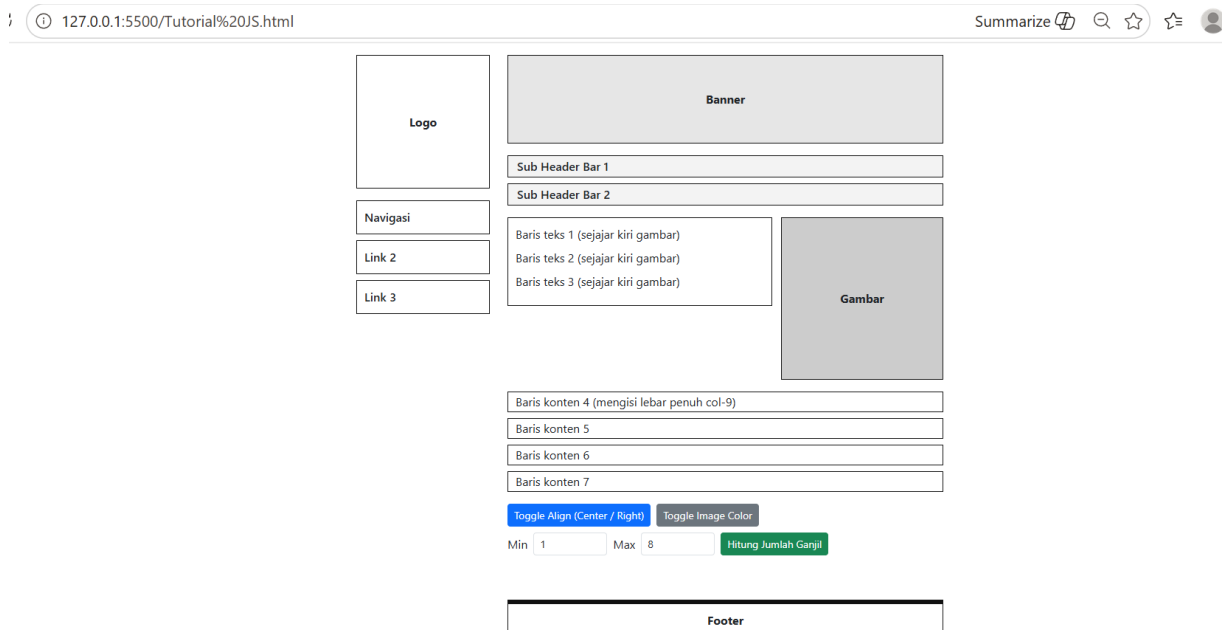
var sum = sumOdd(min, max);
var a = Math.min(min, max);
var b = Math.max(min, max);
var firstOdd = (a % 2 !== 0) ? a : a + 1;
var terms = [];
for (var n = firstOdd; n <= b; n += 2) {
    terms.push(n);
}

resultArea.textContent = 'Hasil: ' + sum + ' (' +
terms.join('+') + ')';

});
})();

</script>
</body>
</html>
```

SS Hasil:



Penjelasan:

Code tersebut adalah penerapan Event-driven programming, di mana sebuah potongan kode JavaScript akan berjalan sebagai respons terhadap interaksi pengguna pada halaman web. Secara teknis, fungsi kalkulator tersebut menggunakan struktur perulangan `for` untuk menelusuri angka dari batas minimal ke maksimal, lalu menggunakan logika kondisional `if` untuk menyaring dan menjumlahkan angka yang hanya bernilai ganjil. Program ini memanfaatkan Document Object Model (DOM) melalui perintah `document.getElementById` untuk mengambil nilai dari kotak input serta memperbarui konten elemen secara dinamis menggunakan properti `.innerHTML` atau `.value`. Selain aspek logika, fitur perubahan tampilan seperti perataan teks dan warna latar belakang bekerja dengan cara memanipulasi atribut gaya atau *class* CSS pada objek elemen HTML tepat saat *event* `onclick` dipicu oleh pengguna.